California Polytechnic State University – San Luis Obispo

F56: Rocket Thrust Vector Control

Team VECTOR

Scope of Work (SOW)

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Abstract

This Scope of Work (SOW) report compiles relevant information regarding the Rocket Thrust Vector Control senior project as sponsored by Ethan Anderson from Cal Poly Space Systems (CPSS). Alongside presenting our core problem statement and project scope, this report also includes stakeholder wants and needs and background research into existing solutions and technical challenges. Furthermore, this document presents key project management-resources like a Gantt chart and Quality Function Deployment (QFD) devised from this information that lay out a roadmap of major milestones along the path towards completion of the project.

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1 Introduction

We are a team of Mechanical Engineering students studying at California Polytechnic State University – San Luis Obispo, and we are developing a rocket thrust vector control (TVC) mechanism for the upcoming Cal Poly Space Systems 500-lb rocket engine.

Our roles in the team are as follows: Ethan is our project sponsor as well as manufacturing and automation engineer in the team, Sahil is a design engineer and technical writer, and Yiming is an analysis engineer and technical writer.

A TVC mechanism allows engineers to guide and control the attitude of a rocket during powered flight through adjustment of the thrust vector. In application, a TVC system is versatile; it can be used to adjust flight path, correct deviations in attitude, and help to align a misaligned thrust vector (with the rocket's center of gravity). One example of such a mechanism is one in which the engine provides thrust while the control system pivots the engine on a gimbal to control the direction of thrust. Our goal for this project is to identify a functional concept, develop detail designs, carry out structural and thermal analysis for, design a robust control system for, and manufacture and integrate a thrust vector control mechanism. Ideally, we will test our system's functionality in Spring 2024 on the engine and test stand as developed by CPSS as part of their engine fire test in order to compare actual results to our expected specifications and verify the success of our project (Cal Poly Space Systems).

2 Background

We have comprehensively documented in the background section key information about our primary stakeholder Cal Poly Space Systems and their specific needs and wants for the project. Furthermore, we have compiled our background research on rocket thrust vector control.

2.1 Stakeholders and Needs

Cal Poly Space Systems is our primary stakeholder for this Rocket Thrust Vector Control project. They have identified the following high-level needs and wants for our project, which have been documented below in Table 1:

Table 1. Preliminary customer needs and wants.

Needs	Wants
Flexible coupling between engine and test stand.	Low cost.
Actuation method for moving engine.	Lightweight.
Integrate with existing hardware.	High range of gimbal angles.
Interface with existing data system.	High gimbal angular velocity.
Structurally rigid interface.	

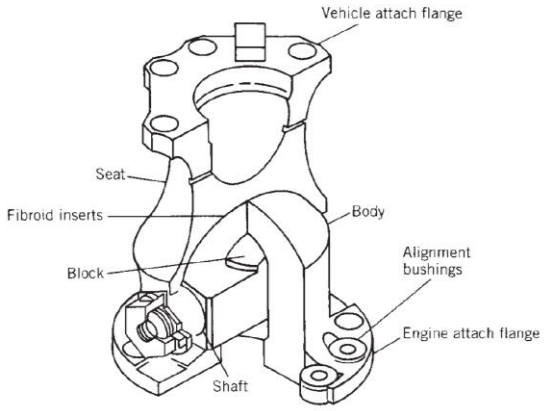
2.2 Existing Products & Solutions

The primary and most apparent challenge posed by this project will be the mechanism through which we will adjust our thrust vector. The textbook Rocket Propulsion Elements breaks down existing solutions for TVC mechanisms into four different types [2]:

1. Mechanical deflection of the main nozzle
2. Deflection of exhaust gas flow using heat-resistant, movable bodies
3. Distortion of exhaust gas flow using fluid injection into the side of diverging nozzle section
4. Separate thrust-producing mechanisms

2.2.1 Pivoted Nozzle

The first type involves pivoting the entire main nozzle on a bearing to point the thrust vector. Notably, this method requires large actuators and power supply as well as a bearing capable of supporting the weight of the engine and transmitting the thrust force. Additionally, Rocket Propulsion Elements identifies the pivoted engine approach as the most efficient (fewest losses), as, unlike the other methods, pivoting the engine does not reduce net thrust or specific impulse of the fuel. For two or more degrees of rotational freedom, a gimbal bearing is required; this is a general term for a mechanism that permits rotation about more than one axis (a mechanism for a single axis would be a hinge). Pictured below in Figure 1 is an example of a gimbal bearing on the space shuttle and example tabulated requirements for the specified gimbal bearing.



(a)

TABLE 18–2. Characteristics and Performance Requirements of the Gimbal Bearing Assembly of the Space Shuttle Main Engine

Engine weight to be supported (lbf)	Approx. 7000
Thrust to be transmitted, (lbf)	512,000
Gimbal assembly weight (lbf)	105
Material is titanium alloy	6Al-6V-2Sn
Dimensions (approximate) (in.)	11 dia. × 14
Angular motion (deg)	
Operational requirement (max.)	±10.5
Snubbing allowance in actuators	0.5
Angular alignment	0.5
Gimbal attach point tolerance	0.7
Overtravel vector adjustment	0.1
Maximum angular capability	±12.5
Angular acceleration (max.) (rad/sec^2)	30
Angular velocity (max.) (deg/sec)	20
Angular velocity (min.) (deg/sec)	10
Lateral adjustment (in.)	±0.25
Gimbal duty cycle about each axis	
Number of operational cycles to 10.5°	200
Nonoperational cycles to 10.5°	1400
Coefficient of friction (over a temperature range of 88–340 K)	0.01–0.2

Source: Courtesy of Aerojet-Rocketdyne.

(b)

Figure 1. Schematic (a) and performance characteristics (b) of Space Shuttle main engine Gimbal Bearing Assembly [2].

Figure 2 depicts three types of gimbaling configurations used in the industry. Force may be transferred from the engine to the payload through a bearing or through fixed length struts as shown below.

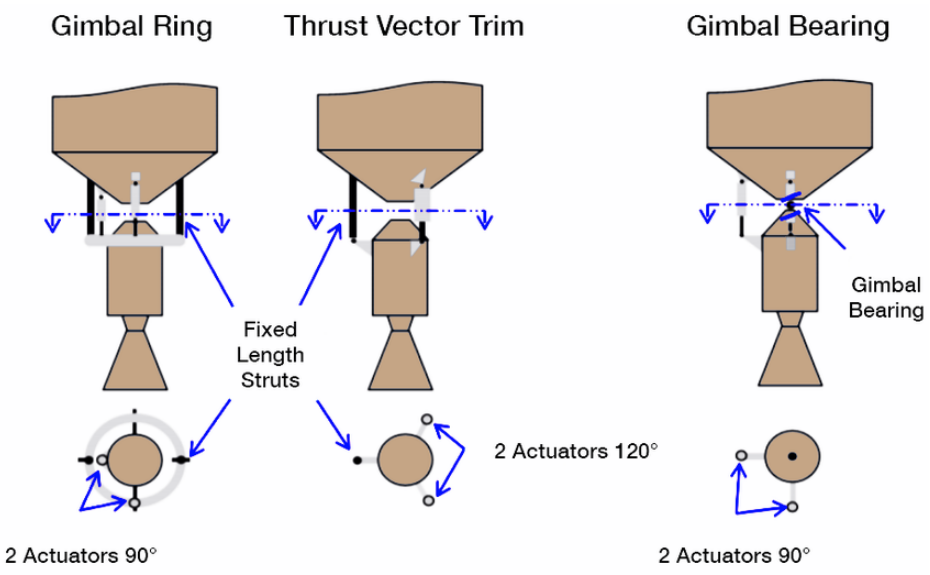


Figure 2. Three actuator support configurations for gimbaling the engine [3].

The gimbal ring would require higher strength actuators. The force the actuators need to provide for the gimbal bearing is a function of the steering angle. The actuators will need to provide a static force of ¼ the total thrust with the engine in the neutral position. Most rockets employ the gimbal bearing configuration because the slew rate requirement centered around 0 force.

2.2.2 Flow Deflection with Mechanically Actuated Bodies

The second type of TVC mechanism involves actuating a mechanical body to deflect the exhaust gas flow and impart a moment upon the rocket. For liquid propellant engines, these bodies are often jet vanes or jet tabs, heat-resistant wing-shaped structures that are actuated into the exhaust flow. Jet vanes are usually placed in line with and inside the exhaust flow and rotated to align the vane normal vector with the flow direction vector. Jet tabs are usually outside of flow and rotated into the flow. Figure 3 is a visual representation of these two mechanisms and their differences. These bodies can erode as they are struck by exhaust, limiting the maximum achievable duration of TVC. Notably, this type of mechanism is characterized by a low actuator power requirement and high slew rate.

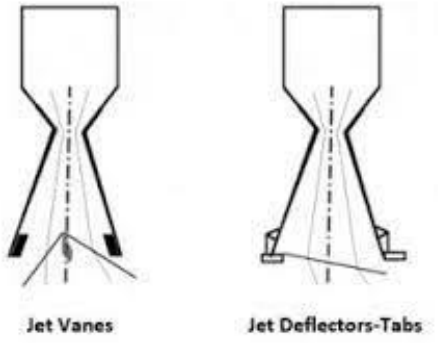


Figure 3. Cross-sectional model comparing jet vanes to jet tabs [4].

2.2.3 Flow Deflection with Fluid Injection

The third type of mechanism involves injecting a liquid or hot gas into the side of the diverging section of the nozzle to distort the flow of the exhaust, redirecting the thrust vector. While liquid injection can achieve a high slew rate and mitigate weight penalty, the liquid tend to be toxic and require specialized maintenance, packaging, and piping. Hot gas injection on the other hand, while lightweight and nontoxic, requires special valves and contacts designed for hot gas. Like the jet vane mechanism, this method also has limited duration.

2.2.4 Auxiliary Thrust Chambers

The fourth type of mechanism involves separate thrust-producing mechanisms; for liquid propellant engines, these are either hinged/pivoted auxiliary thrust chambers fed from the main system or a turbine exhaust gas swivel. Both methods, notably, are limited in the moments and forces that they can apply to a rocket. They also add significant complexity and failure modes in the system that can be undesirable.

2.2.5 Various Existing Solutions

One specific solution we decided to research was from the National Aeronautics and Space Administration (NASA), as they host an extensive knowledge base of documentation online. NASA manuscript AAS 23-152 “Overview of the SLS Core Stage Thrust Vector Control System Design” highlights the TVC system design for the Space Launch System (SLS) rocket, designed for the Artemis space program [5]. They divide their system into four main assemblies: the TVC actuators, the actuator controllers, the Core Auxiliary Power Units (CAPU), and CAPU controllers. Each of the four Aerojet Rocketdyne RS-25 engines has two mounted hydraulic actuators with one controlling the engine’s pitch and the other controlling the yaw. The power for these actuators is delivered by the CAPU hydraulic system and both the hydraulic system and actuators have closed-loop feedback for valves, pressure differentials, and loads. Feedback monitoring flow indicates feedback for position positions and valve positions. The hydraulic system includes a hydraulic pump, reservoir, flow manifold, and heat exchangers.

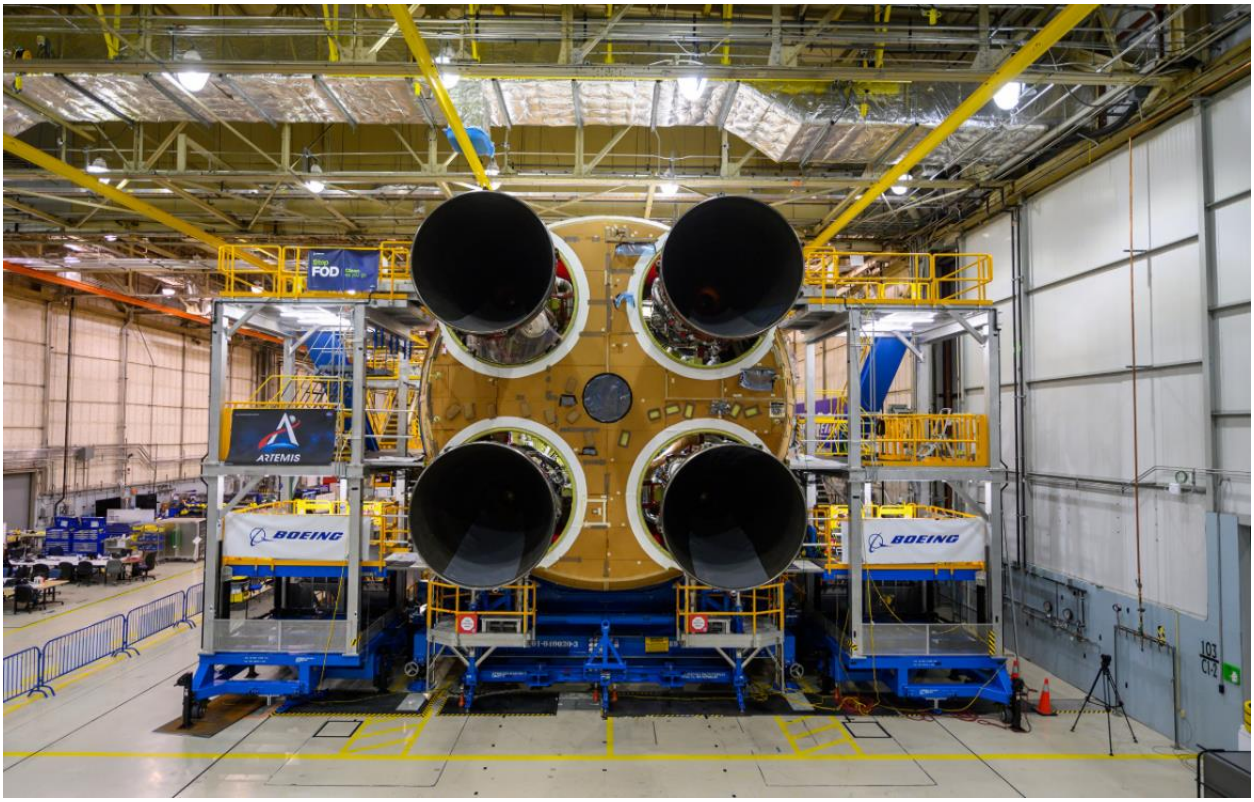


Figure 4. SLS Core Stage with four RS-25 engines installed [6].

While thrust vector control actuation systems have commonly been hydraulics-powered, there are also electromechanical actuation methods, which the 1998 publication "Modeling and Estimation for Electromechanical Thrust Vector Control of Rocket Engines" describes as having the potential for "increased reliability, reduced weight and volume of the end-to-end power and actuation system, and decreased maintenance requirements." [7]. The NASA-supported publication models the system (see below) dynamics of a motor-driven electromechanical actuation system for gimbaled engine TVC in which a motor torque input can be processed into transmitted force and displacement outputs.

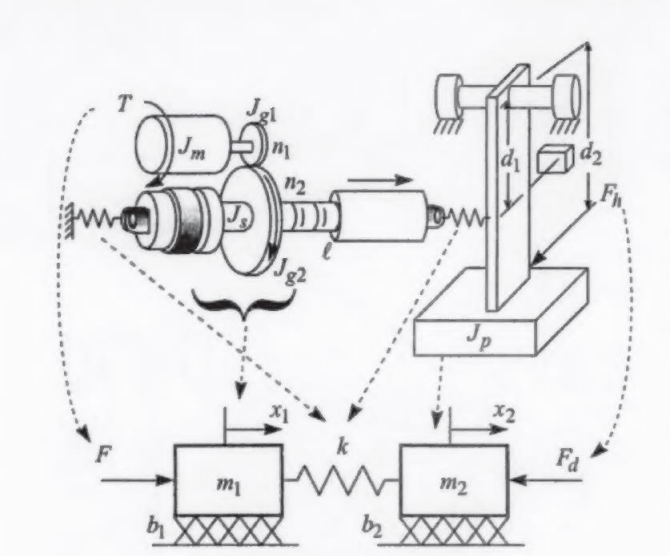


Figure 5. Representative model for Electromechanical TVC system [7].

One successful example of a electromechanical TVC system in production is that of SpaceX. While the company tends prefers to keep its proprietary information and knowledge secret, the Raptor engines for the SpaceX Super Heavy rocket booster use "fully electric motors" and has "fewer points of failure and is significantly more energy efficient than traditional hydraulic systems" [8].

According to the 1974 case file "Solid Rocket Thrust Vector Control", NASA discusses actuator selection according to the various torques that are experienced by them [9]. The actuation torque is the summation of all the contributing torques. They use the maximum actuation torques to determine the total required torque activation. One torque that is included in this actuator torque is joint spring torque. Joint spring torque is the maximum value of shear modulus that is used in the design of the actuator. In the case file, NASA asserts that if the joint is to be vectored to different angles during motor operation take advantage of the reduction in spring torque due to motor pressure to reduce the actuation power requirements. It is desirable that the minimum spring torque be sufficiently large to prevent negative joint spring stiffness due to pressure. NASA also mentions a variety of other torques that they consider when specifying their linear actuators such as offset torque, inertial torque, gravitational torque, insulating boot torque, internal aerodynamic torque, external aerodynamic torque, etc. In future research, we will study the cause and effects of all these torques to select the actuators efficiently.

According to Ultra Motion, their B1 actuators can provide forces up to 500 lbf with speeds up to 2.75 in/s [10]. These actuators are something we should conduct much more detailed research on as they could be similar to what we would need.

2.3 Technical Challenges

Separate from the technical challenges posed by the actual thrust vector controlling mechanism classified as "Existing Products and Solutions", we also face a breadth of challenges throughout the design and analysis process of our system. NASA provides many standards to guide this process; a selection of the standards we have opted to reference along with a few key requirements in each are as follows:

- NASA-STD-5001 *"Structural Design and Test Factors of Safety for Spaceflight Hardware"* [11]
    - [FSR 13] When using the prototype structural verification approach, metallic structures shall be verified to have no detrimental yielding at yield design load before testing to full qualification load levels.
    - [FSR 15] Analysis shall be provided with an acceptable engineering rationale for the "no-test" option.
    - [FSR 18] The design factors of safety and test factors of this NASA Technical Standard are the minimum required values for NASA spaceflight structures and shall be applied to the limit stress condition, including additive thermal or pressure stresses.
        - SFy = 2.0, SFu = 2.6
- NASA-STD-5002 *"Load Analyses of Spacecraft and Payloads"* [12]
    - [LAR 1] A minimum of three load analysis cycles shall be performed: Preliminary Design Load Cycle (PDLC), Final Design Load Cycle (FDLC), and Verification Load Cycle (VLC).
    - [LAR 5] Limit loads shall be determined that will not be exceeded with 99.87/50 (99.87 percent probability at 50 percent confidence level).
    - [LAR 9] For random vibration loads analysis, peak responses shall be used for the limit loads.
    - [LAR 13] To initiate the design process, a set of preliminary limit loads shall be developed and used for the sizing of primary structure, secondary structure, and components.
- NASA-STD-5017 *"Design and Development Requirements for Mechanisms"* [13]
    - Torque Margin / Force Margin is the excess capability of a mechanism to impart, maintain, or prevent motion, or to provide any required acceleration (for unforeseen events and effects)

$$\boldsymbol{torque\ margin} = \frac{T_{avail}}{\sum FS_f T_f + \sum FS_v T_v + \sum FS_a T_a} - \mathbf{1}$$

    - [DDMR 9] All torque margin requirements shall be applied under worst-case conditions throughout the mechanism's life, including throughout life testing.

These standards, though they vary in scope, specify operational requirements and inform design and analysis decisions that we expect to face throughout our senior project design process. In addition to these, there are also standards for materials and processes and material fracture control [14] [15].

Design of Liquid Propellant Rocket Engines by Huzel and Huang lists four engines with different thrust and gimballing characteristics [16]. These specifications helped us to determine adequate target specifications for our own system.

- A-1 Stage Engine, Thrust 750,000 lb
    - Minimum Acceleration 1 rad/sec^2
    - Maximum Velocity 10 deg/sec
    - Displacement 14deg
- A-2 Stage Engine, Thrust 149,500 lb
    - Minimum Acceleration 2 rad/sec^2

    - Maximum Velocity 15 deg/sec
    - Displacement 6 deg
- A-3 Stage Engine, Thrust 16,000 lb
    - Minimum Acceleration 2 rad/sec^2
    - Maximum Velocity 15 deg/sec
    - Displacement 7 deg
- A-4 Stage Engine, Thrust 7500 lb
    - Minimum Acceleration 2 rad/sec^2
    - Maximum Velocity 15 deg/sec
    - Displacement 7 deg

Huzel and Huang also go on to list primary design considerations:

1. Required pivotal movement of engine assembly or thrust chamber for thrust vector control. - Generally ranging from ±4° to ±10°.
2. Required adjustment for thrust alignment and positioning.
3. Thrust Level. - This determines the structural and bearing design of the gimbal mounts.
4. Required operational life. - Generally 1000 cycles minimum.
5. Minimum deformation of the bearing surface, prevention of bearing surface galling.
6. Propellant duct installation. - Some designs require flow of one of the propellants through the center of the gimbal mount.
7. Maintenance of the gimbal mount. - This is largely affected by the lubrication requirement of the bearing surfaces. If possible, a gimbal boot should be provided to protect the bearing surfaces from dust, water, and foreign materials.
8. Lightweight. - High strength-to-weight ratio materials should be used.

In terms of actuator positioning with respect to the engine, a balance must be struck between long-stroke and short-stroke positions [18] [20]. Long stroke actuator results in a low effective gimbaled mass resulting in smaller actuators, but the spring rate is low and so is the stiffness. This being the inverse for a short stroke. The two configurations are shown below.

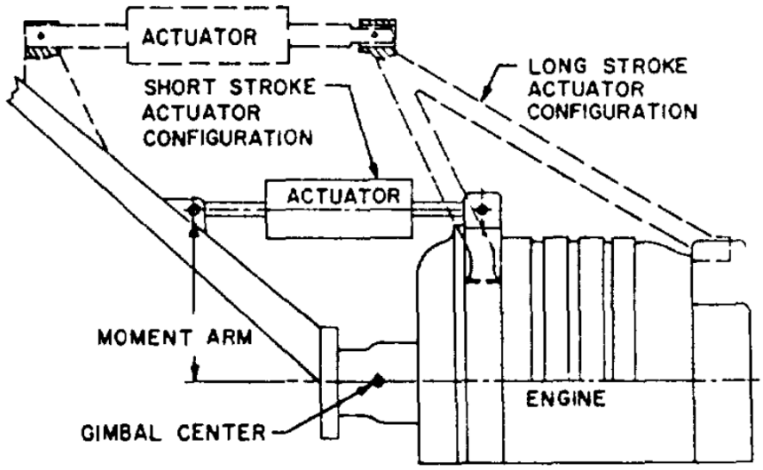


Figure 6. Engine actuator installation configurations.

Another technical challenge will be crosstalk [19]. Assuming a linear actuation method is used, crosstalk is when the motion of an actuator in one plane affects the other actuator in its plane. This must be minimized to achieve high positional resolution and accuracy.

The last technical challenge will be the control system. This system will likely consist of actuators controlled by commands which take into account the vehicles deviation from being upright and it's response to corrective stabilizing action. A diagram of this system is shown below. The system will need to sense the angle of the engine, the angular rate of the engine, and convert these measurements to an output signal to send to the actuators. Malfunction safety circuits will also need to be included to effect engine cutoff in case of actuator failure or erratic operation [17].

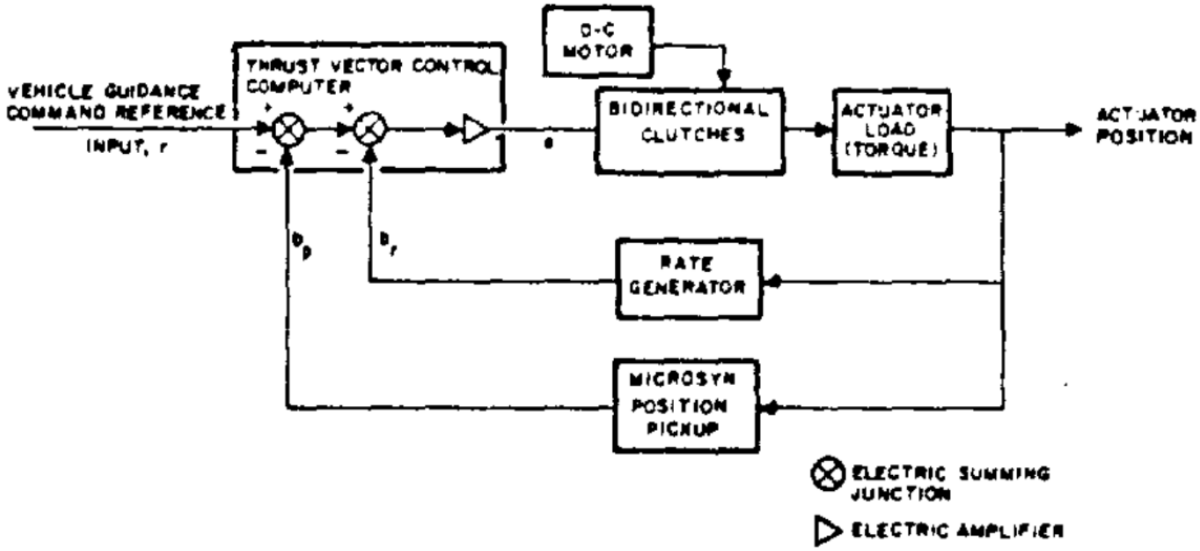


Figure 7. Block diagram for a thrust vector control system using electromechanical actuators

3 Project Scope

3.1 Boundary Sketch

Below in Figure 8 is a rough boundary sketch that visualizes the position and interfaces of the mechanism that we are designing in context with the rocket engine and test stand. In a boundary sketch, areas in which the dotted boundary line intersects existing structure indicate interfaces that we may need to design our mechanism for. The two key interfaces here are the one with the test stand mounting plate and the one with sides of the engine; the interface pictured is oversized, as to not over-constrain the possible positions for actuator mounting.

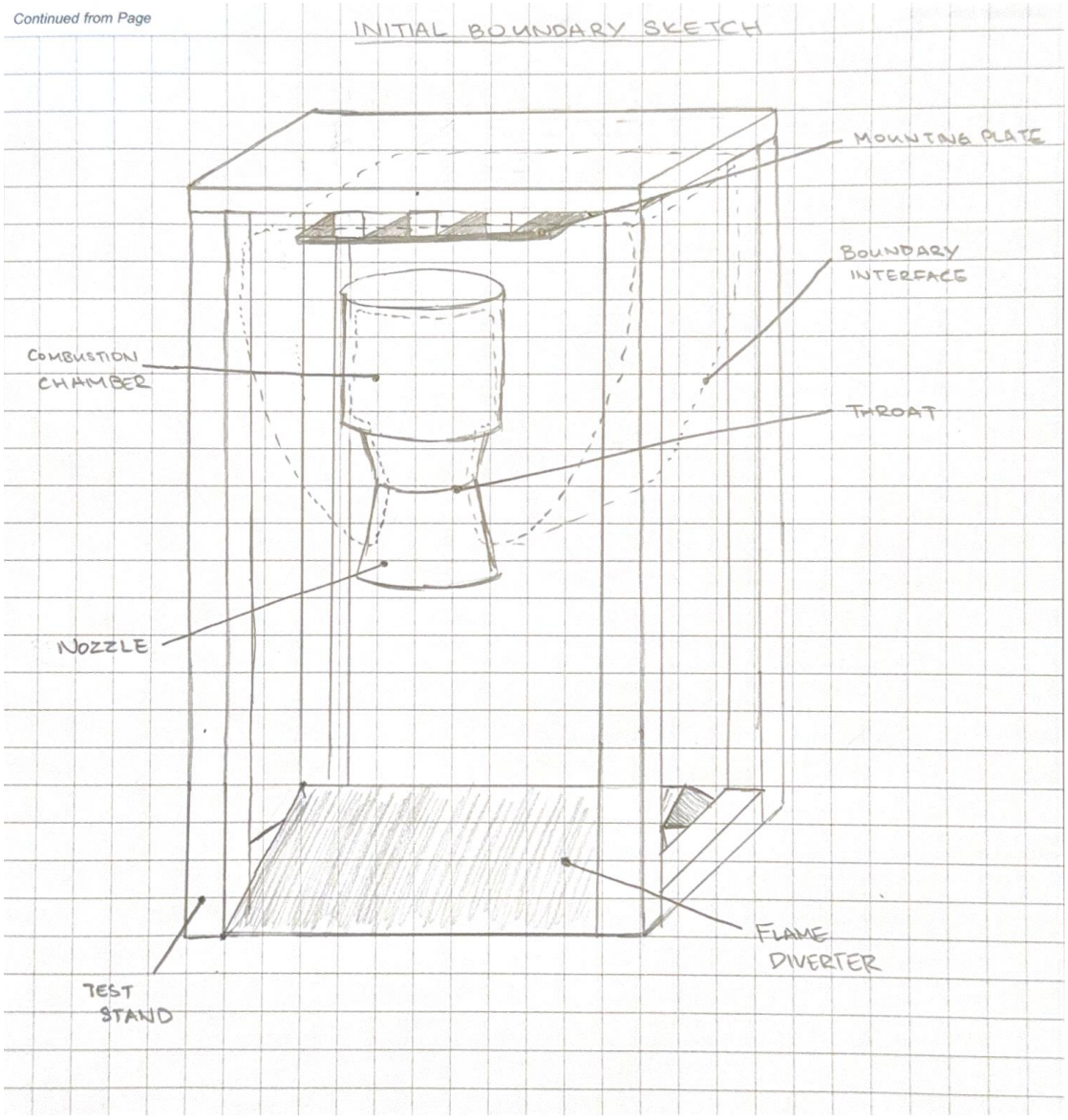


Figure 8. Boundary Sketch for Rocket Thrust Vector Control project

3.2 Functional Decomposition

To best map out our goals for this project and for the mechanism, we have created a functional decomposition. The functional decomposition is a visual representation and hierarchy of all the functions that our TVC mechanism should possess. Broader functions toward the top of the "tree" are broken down into more specific functions that, eventually, we will seek to ideate upon one at a time.

Below in Figure 9 is the functional decomposition for our TVC mechanism. The diagram includes core mounting, load bearing, and control functions. Notably, we divided the mechanism's control into three subfunctions: accepting input signals, outputting feedback signals, and adjusting the momentum of the exhaust flow. For context, the control system that we develop will need to interface between the TVC actuators and CPSS's Data Acquisition (DAQ) system [1]. This means that all input and output communication (including commands and feedback) will pass through our system. Furthermore, we describe the method of thrust vector control as "adjusting the momentum of the exhaust flow" as, from a dynamic perspective, all methods of TVC involve the adjustment of exhaust flow momentum.

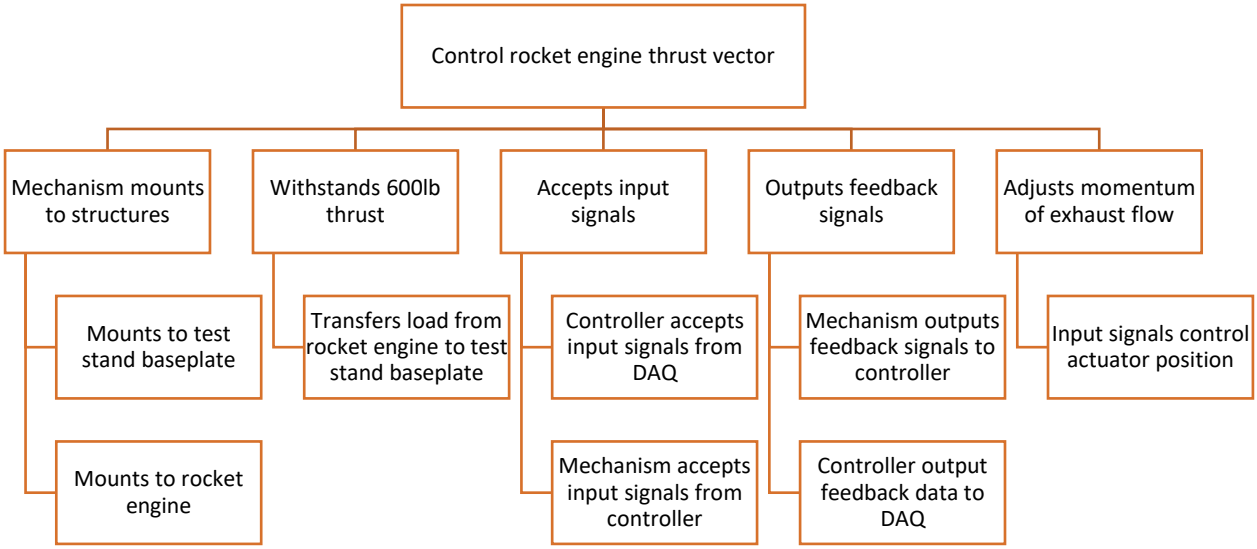


Figure 9. Functional Decomposition for Rocket Thrust Vector Control senior project

3.3 Project Deliverables

At the end of this project, we plan to deliver the following components to our sponsor:

- gimbal to allow rotational freedom of the engine in two directions
- actuation method to control said rotation
- relevant mounts and mounting hardware
- electronic control system.

In terms of performance, we plan to deliver the following:

- 6 degrees of motion in both directions
- 16 degrees per second of angular rate
- angular precision and accuracy of ±1°

4 Objectives

4.1 Problem Statement

Cal Poly Space Systems (CPSS) is currently in the process of developing a 500-lb actively stabilized rocket engine. In preparation for future missions, CPSS is seeking the development and integration of a Thrust Vector Control (TVC) system, which will be essential for enabling precise control of the rocket's thrust vector and allowing for in-flight attitude adjustments and propulsive landing capabilities. The project requires the design and specification of actuators, a 2-degree-of-freedom gimbaling mechanism, and the corresponding mounting hardware. The successful implementation of this TVC system is pivotal in ensuring the rocket's stability and maneuverability for future missions.

4.2 Quality Function Deployment

In our Quality Function Deployment (QFD), we broke down the main customer, Cal Poly Space Systems, into four main customers for the Thrust Vector Control System:

1. **CPSS Test Engineers**, concerned primarily with the high-level requirements for performance, integration, and cost for the mechanism
2. **CPSS Design**, concerned mainly with the flexibility of the mechanical and controls system design with future rocket engines
3. **Manufacturing**, concerned with the manufacturability of the mechanism’s components with limited machines and tooling
4. **Assembly & Integration**, concerned with the mounting interfaces for the mechanism with the engine and test stand baseplate

After creating requirements encompassing mechanical properties, mechanism design, system performance, manufacturing, and assembly. Weights for each requirement were assigned according to each of the customer’s concerns as described above. For the current products section, we identified three primary “competitor” mechanisms as produced by:

1. **SLO Propulsion Technologies**, a Cal Poly SLO club developing small-scale liquid propulsion engines with regenerative cooling and TVC mechanisms.
2. **National Aeronautics and Space Administration (NASA)**, the United States independent government agency leading the civil space program and space research.
3. **Space Exploration Technologies Corp. (SpaceX)**, the American aerospace company designing and manufacturing spacecraft and providing launch services, defense solutions, and satellite communications.

However, due to limited available information and immediate availability on these current technologies, we have opted not to score the current products according to our requirements and target values (for which

the scale of NASA and SpaceX's products would far exceed regardless). Instead, the mechanisms and limited documentation provided by each of these "competitors" will serve as benchmarking for our background research into TVC mechanisms. Finally, we devised a list of specifications from our requirements encompassing many of the same system criteria before identifying relationships and directions of improvement for each. These specifications reflected the requirements for our system, defined using quantifiable metrics to better allow for judgment of success.

4.3 Specifications

Table 2. Specifications table for Rocket Thrust Vector Control project, as derived from QFD

Number	Specification	Type of Target	Target Value	Compliance	Risk
1	Weight	Minimize	50 lbs	A, I	Low
2	Cost	Minimize	$667	A, I	Hi
3	Angular Velocity	Target	16°/s	A, T	Hi
4	Maximum Load	Target	600 lbs	A	Low
5	Range of Gimbal Angle	Maximize	±8°	A, I	Low
6	Designed Factors of Safety	Target	Roughly 1.5-3	A	Low
7	Natural Frequency	Maximize	40Hz	A	Med
8	Positional Accuracy & Precision	Maximize	Nominal within ±1°	T	Med
9	Safety	Maximize	Conforms to CPSS Standards	I	Low

To break down the specifications listed in Table 2 above:

Weight shall be minimized when possible; any extra weight added to the rocket incurs a penalty or loss of fuel efficiency that reduces the maximum flight time and distance attainable by the rocket [2].

Cost shall be minimized when possible and shall not exceed our project budget.

Angular velocity shall meet a targeted value of 16 degrees per second. While a high angular velocity allows our mechanism to execute more complex maneuvers, certain angular velocities only need to be met. Exceeding the target value yields minimal benefit for increased investment.

Maximum load shall meet a target value of 600 pounds (not counting our factor of safety) to ensure the system does not fail structurally.

Gimbal angle range shall be maximized when possible; a larger range of angles allows for larger applied moment and, thus, faster maneuvers.

Desired factors of safety shall always be targeted to balance safety and margin for our system with weight and cost. Design standards and analysis shall inform our factor of safety requirements for each component.

Natural frequency shall be maximized when possible; rocket engines produce significant, high-frequency vibrational environments during launch that can cause resonance failure in low natural frequency components and assemblies [2].

Positional accuracy and precision shall be maximized when possible. Our physical system should match our controls models and calculations as closely as possible to avoid critical failure during firing.

Safety shall be maximized in accordance with CPSS standards for safety [1].

5 Project Management

Once our team was formed, the first step was to set up a file sharing and communication method. We decided to use Microsoft teams and Jira software. Next, we formulated a team contract to fully understand what roles each group member would be responsible for. We then created a project plan which consisted of all the major milestones. Our final project plan can be seen in the form of a Gantt Chart in Appendix B of the scope of work. The steps we followed to create the Gantt chart were:

1. List all project milestones
2. For each of the milestones above write down the necessary steps to complete them
3. Make all the tasks S.M.A.R.T (Specific, Measurable, Attainable, Realistic, Timely)
4. Decide the timeframe for each task and write down the due dates.
5. Arrange the tasks in the best order to efficiently complete them
6. Lastly, decide the team member who will lead each task.

We decided during the first quarter we would devote most of our time to research. As we compile research, we are consciously thinking about which design and analysis tools we would need. We are likely to use a variety of hand calculations to specify our actuators, mounting systems, and our gimbaling method. We will need to model everything with AutoCAD and use finite element analysis. We will also need to perform some vibration analysis and heat transfer analysis for the mounting hardware.

Table 3 below shows the project milestones and the assigned due date.

Table 3. Key senior project milestones and corresponding due dates.

Project Milestones	Due Date
Scope of Work	October 23, 2023
Preliminary Design Review	November 16, 2023
Interim Design Review	January 23, 2024
Critical Design Review	February 8, 2024
Manufacturing and Test Update	March 14, 2024
Senior Exam	April 4, 2024
Expo	May 31, 2024
Final Design Review	June 7, 2024

Looking at the table above, the next step after completing the scope of work is to start the preliminary design review. To effectively complete the preliminary design review, we have a series of tasks that we need to accomplish. The first step we will take is to do more ideation and complete an ideation model. An ideation model is a concept model that is quick to make, allowing us to visualize and share ideas, perform preliminary tests, and develop new perspectives. After forming our different ideation models, we will discuss the advantages and disadvantages of each model before refining our ideas and selecting a concept. After selecting the concept, we will build a model. A concept model is a much more realistic representation of our design, and it will show some functionality. The prototype should not take more than 2 days to build. Then, we will present our concept prototype along with our preliminary design report.

Following PDR, we will create a full detail design and set of engineering drawings for Critical Design Review (CDR); the completion of CDR will mark the beginning of manufacturing and assembly of our system. Once our system is complete, we will test it according to test protocols that we shall draft before reporting on our final findings and successes in our Final Design Review (FDR) and display for the Senior Project Expo.

Finally, it is important that we share here that we will be pursuing grants in order to better fund the material cost for our project. Notably, we will pursue Mechanical Engineering Discretionary-fund Allocation Committee (MEDAC) and Baker-Koob Endowment funding for our project; these are both grant programs provided through Cal Poly.

6 Conclusion

Our project seeks to fulfill the need identified by Cal Poly Space Systems for a thrust vector control (TVC) system for their in-development 500-lb rocket engine. The TVC mechanism will mount to the engine and test stand and be capable of rotation about two axes, with targeted angular position range of ±6° and angular velocity of 16°/s. The success of this project will enable in-flight attitude corrections and propulsive landing capabilities for not only the current rocket engine, but future missions as well.

References

[1] Cal Poly Space Systems, Propulsion Team, *Dulcimer Engine: CDR*, March 2023. PowerPoint Presentation.

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Appendices

Appendix A – QFD House of Quality

Correlations	
Positive	+
Negative	–
No Correlation	

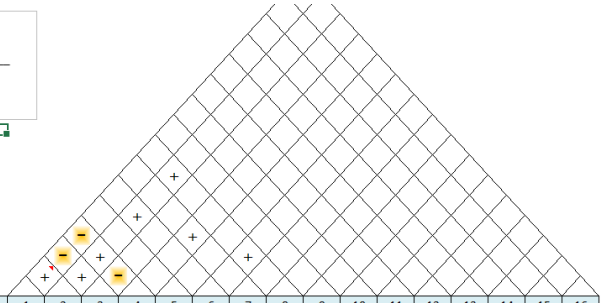
Relationships	
Strong	●
Moderate	○
Weak	▽

Direction of Improvement	
Maximize	▲
Target	◇
Minimize	▼

QFD House of Quality

Project: _____Thrust Vector Control_____

Revision Date: ___10/23/2023___



		WHO: Customers																																
										Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16								
										Direction of Improvement	▼	▼	◇	◇	▲	◇	▲	▲	▲	▲							NOW: Curr. Products							
Row #	Weight Chart	Relative Weight	CPSS Test Engineers	CPSS Design	Manufacturing	Assembly & Integration	Maximum Relationship	WHAT: Customer Requirements (Needs/Wants) / HOW: Engineering Specifications (Tests)	Weight	Cost	Angular Velocity	Maximum Load	Range of Thrust Vector Angle	Design Factor of Safety	Natural Frequency	Safety	Positional Accuracy and Precisio									VECTOR!	Competitor #1: Slo Propulsion T	Competitor #2: NASA	Competitor #3: SpaceX		Row #
1	■	8%	6	2	5	10		Easily Assembled																		3	3	3	1		1
2	■	6%	3	10	1	6		Actuation Method																		5	5	5	5		2
3	■	11%	10	5	6	10		Integrate with existing hardware																							3
4	■	8%	10	5	1	10		Integrate with existing software																							4
5	■	6%	10	5	1	6	9	Structurally rigid	○					●	●	○												5	5		5
6	■	11%	8	8	7	8	9	Low Cost		●																		4	5		6
7	■	7%	8	6	1	8	9	Light Weight	●																			4	5		7
8	■	6%	10	8	1	3	9	High Gimbal Angle					●																		8
9	■	6%	10	8	1	1	9	High Gimbal Angle Rate			●	●																			9
10	■	9%	1	8	10	1		Easily Manufacturable																							10
11	■	6%	10	5	1	6	9	Angular Position Accuracy									●														11
12	■	7%	10	8	1	6	9	Closed-Loop Control System									●														12
13	■	10%	1	10	6	10		Engine Diameter Compatability																							13
14		0%																													14
15		0%																													15
16		0%																													16
								HOW MUCH: Target Values	50 lbs	$667	16 degrees/second	600 lbs	+/- 6 degrees in 2 rotational directions	Roughly 1.5 to 3	TBD	Conform to CPSS Standards	Nominal within +/- 1°													
								Max Relationship	9	9	9	9	9	9	9	3	9														
								Technical Importance Rating	79.5	97.69	51.65	51.65	56.95	57.22	57.22	19.07	122.1	0	0	0	0	0	0	0							
								Relative Weight	13%	16%	9%	9%	10%	10%	10%	3%	21%	0%	0%	0%	0%	0%	0%	0%							
								VECTOR!																							
Curr. Products								itor #1: Slo Propulsion TVC system																							
								Competitor #2: NASA																							
								Competitor #3: SpaceX																							
								0																							
								Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16							

Appendix B – Gantt Chart

